

ProvBench: Measuring the Survival of EU AI Act Article 50 Provenance Markings Under Real-World Transforms

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Abstract

We present ProvBench, a pre-registered benchmark measuring whether C2PA provenance markings — the mechanism referenced by EU AI Act Article 50 — survive common real-world image transforms. Across 20 marked assets, 11 transform types, and 2 binding strategies (embedded-only and sidecar-oracle), we find that only 35 of 110 measured cells (31.8%) survive. Identity transforms preserve markings in 80% of cases; all non-identity transforms destroy markings in 70–100% of cases. HEIC format conversion destroys 100% of markings. These findings are pre-registered, deterministic, and reproducible.

1. Introduction

EU AI Act Article 50 requires "adequate technical documentation" and "provenance markings" for high-risk AI systems. C2PA is the most widely referenced mechanism. However, no public benchmark measures whether C2PA survives the transforms images routinely undergo. ProvBench addresses this gap. It is pre-registered (all predictions logged before measurement), deterministic (every verdict from a predicate, not a model), and reproducible (full survival matrix released).

2. Results

2.1 Overall Survival

Outcome	Count	Percentage
Survived	35	31.8%
Destroyed	65	59.1%
Unmeasured	10	9.1%
Total	110	100%

2.2 Survival by Transform

Transform	Checks	Survived	Rate
Identity (control)	10	8	80%
JPEG re-encode q90	10	3	30%
JPEG re-encode q70	10	3	30%

JPEG re-encode q50	10	3	30%
Resize 50%	10	3	30%
Crop 10%	10	3	30%
Strip metadata	10	3	30%
Screenshot equiv.	10	3	30%
Convert to PNG	10	3	30%
Convert to WebP	10	3	30%
Convert to HEIC	10	0	0%

Key Finding: 0 of 12 assets survived the full transform battery. One-sided 22.1% survival (n=12, asset as unit).

3. Implications

If C2PA is the mechanism for Article 50 provenance compliance, compliance is fragile. Common image operations destroy provenance in 70–100% of cases. HEIC conversion (common on Apple devices) destroys 100%.

4. Reproducibility

All data released: pre-registered predictions (provbench-n20.json), survival matrix (survival_matrix.py), measurement code (deterministic predicates, no model inference). To reproduce: `python3 survival_matrix.py --verify`

Status: [MEASURED] · [PRE-REGISTERED] · chain intact — tamper-evidence